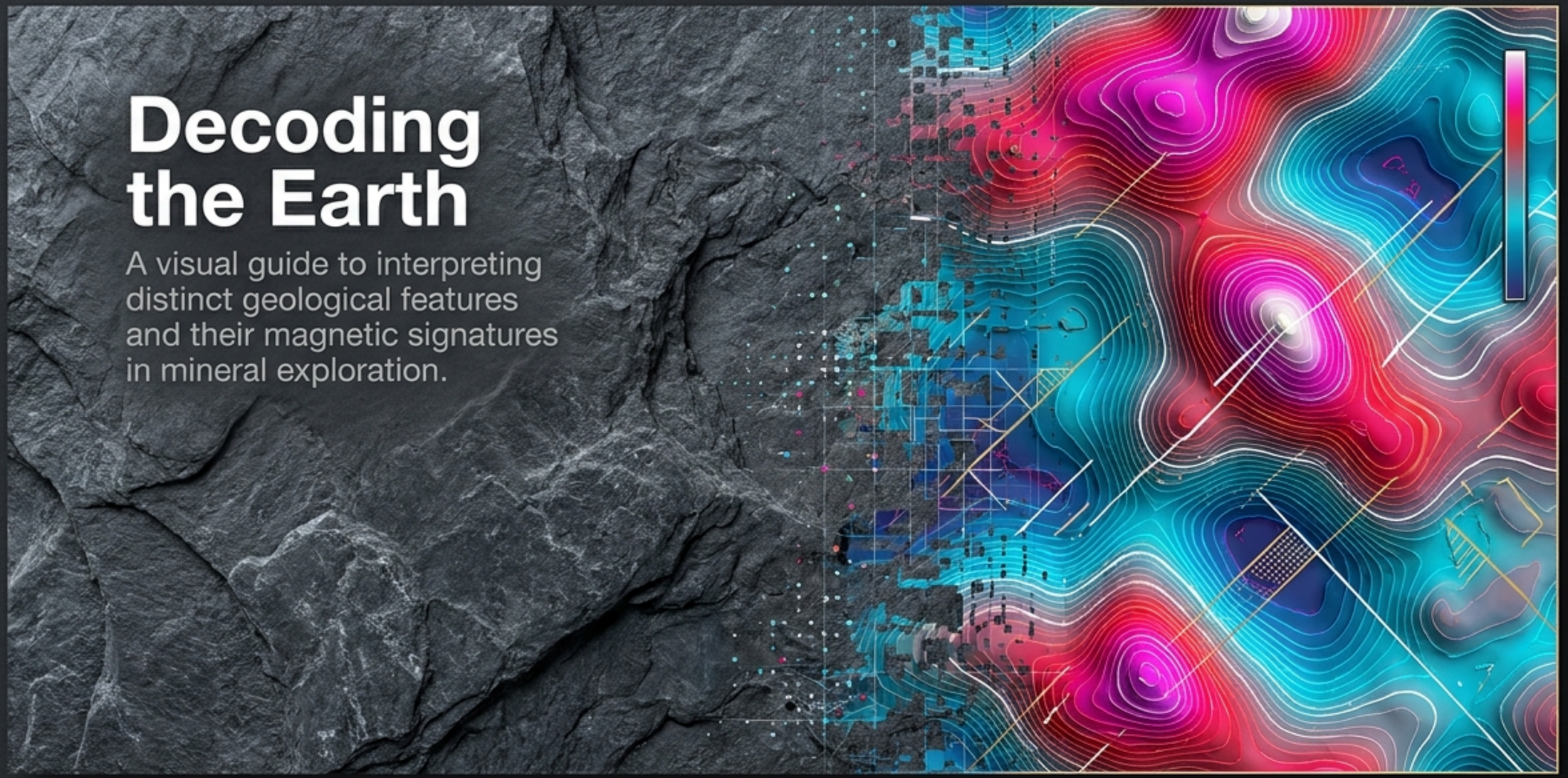
Decoding the Earth

A visual guide to interpreting distinct geological features and their magnetic signatures in mineral exploration.



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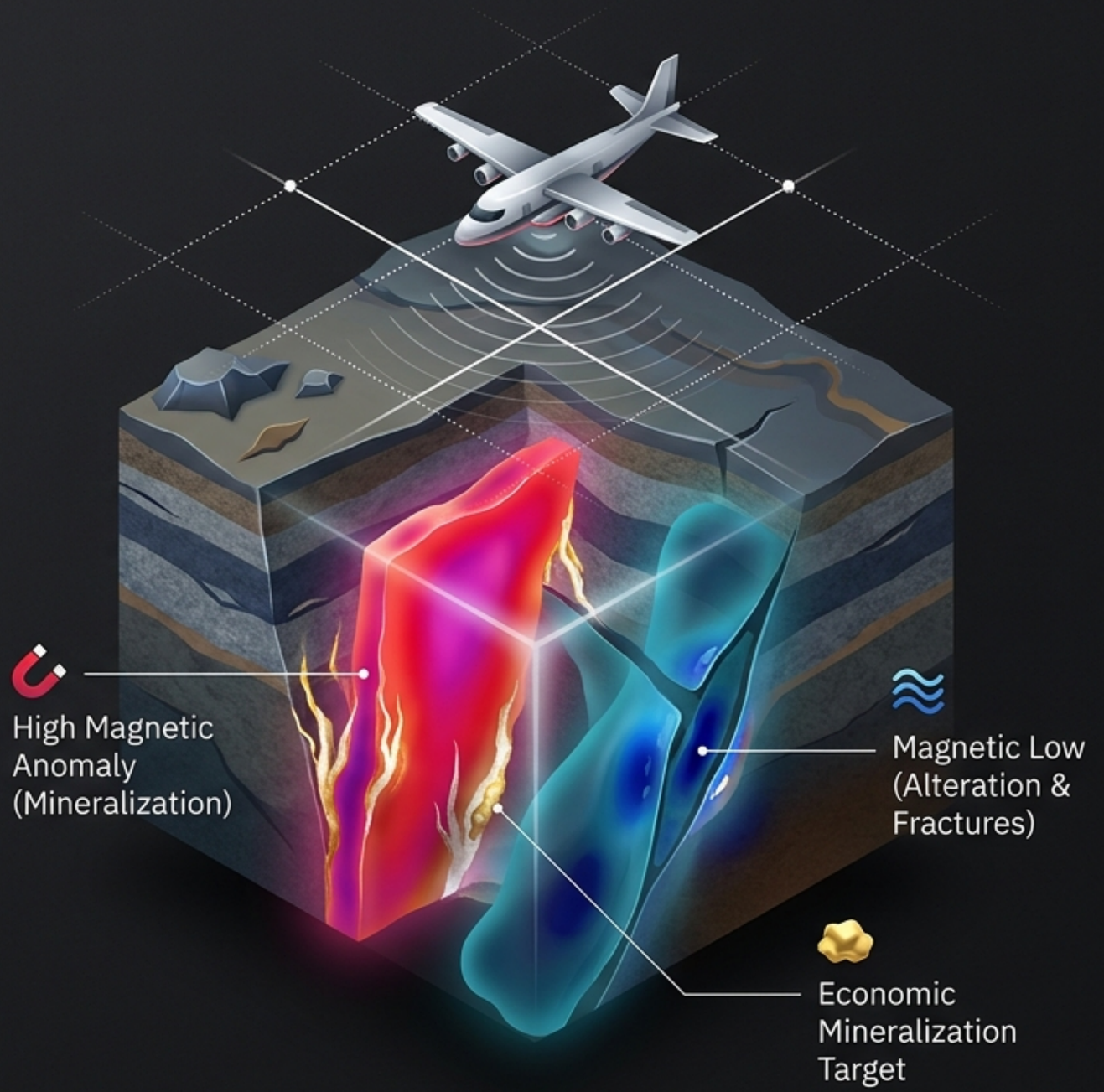
Geophysical surveys are the X-rays of the earth.

To find deep mineral deposits, exploration teams must translate hidden physical geology into readable magnetic signatures.

Subterranean Geology
(The hidden physical reality)

Magnetic Signature
(The geophysical symptom)

Interpretation
(The targeted exploration model)



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The Core Four

These structures and alteration styles produce very different magnetic signatures.
Together, they represent the essential vocabulary for diagnosing subterranean mineral systems.



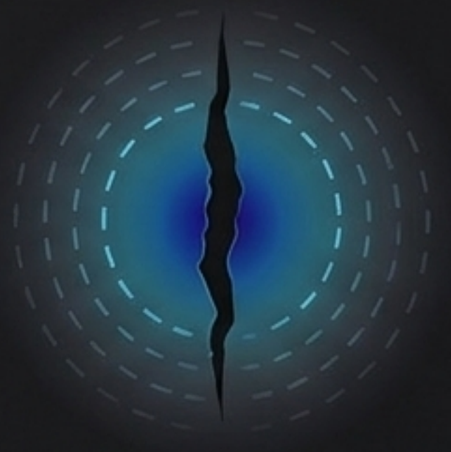
Magnetic Dyke



Fault / Shear Zone



Quartz Vein

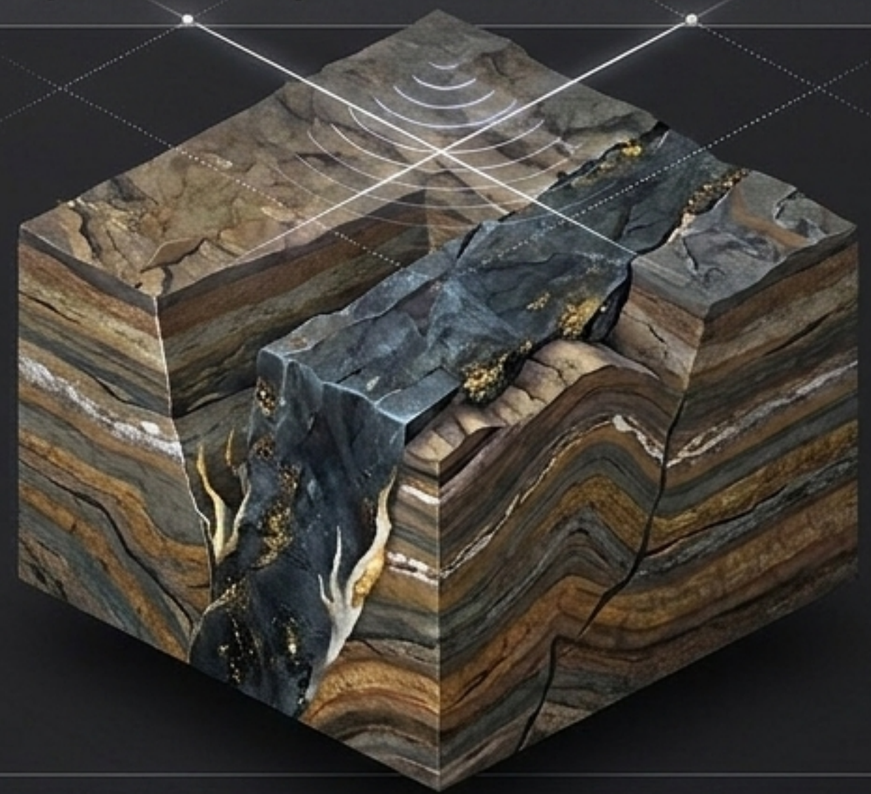


Magnetite Destruction

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Feature 1: Magnetic Dyke

Physical Reality



Magnetic Reality



What it is: Intrusive Igneous Sheet.

Formed when magma (dolerite, basalt, or lamprophyre) intrudes into fractures and cools discordantly across older formations.

Magnetic Expression: Strong Linear High.

Produces a sharp positive magnetic anomaly due to high concentrations of primary iron-titanium oxides like magnetite.

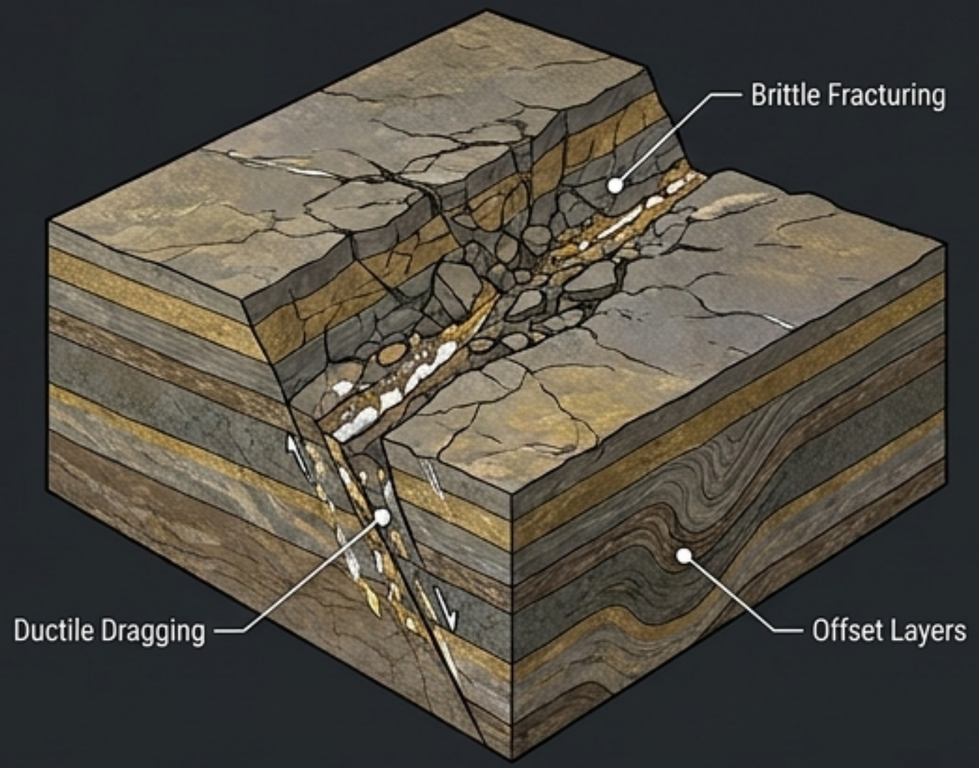
Geological Role: The Structural Barrier.

Acts as a rigid wall, a thermal driver for fluid circulation, and establishes relative chronological timelines.

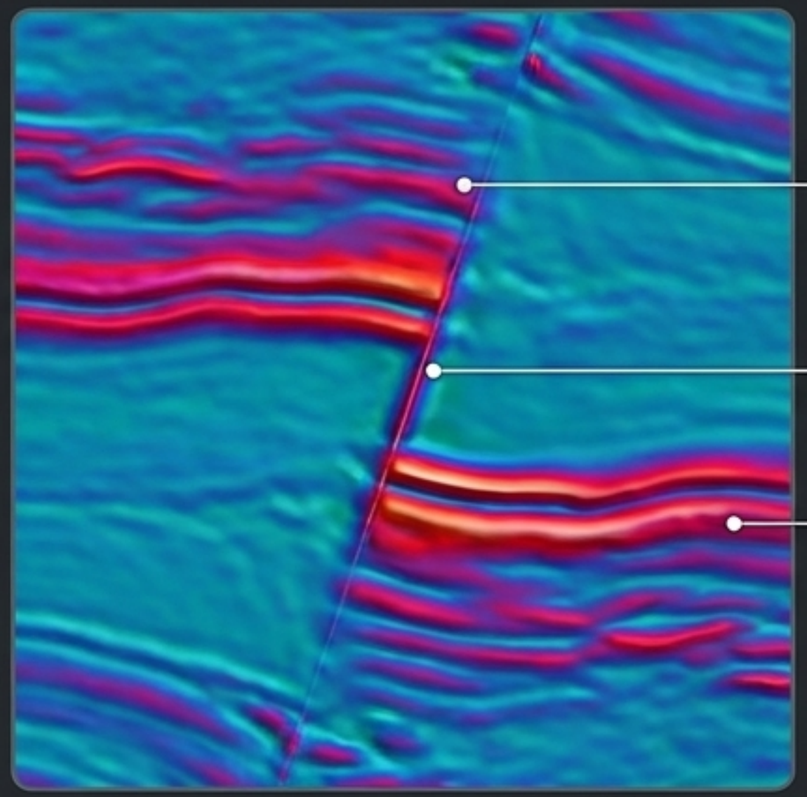
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Feature 2: Fault or Shear Zone

Physical Reality



Magnetic Reality



What it is: Fracture & Strain Zone.

A fracture where blocks of the Earth's crust have moved relative to each other, subjected to intense strain.

Magnetic Expression: Discontinuity & Offset.

Appears as a linear truncation of magnetic trends. (Note: Can manifest as a linear low if the zone has undergone destructive carbonation or sulfidation).

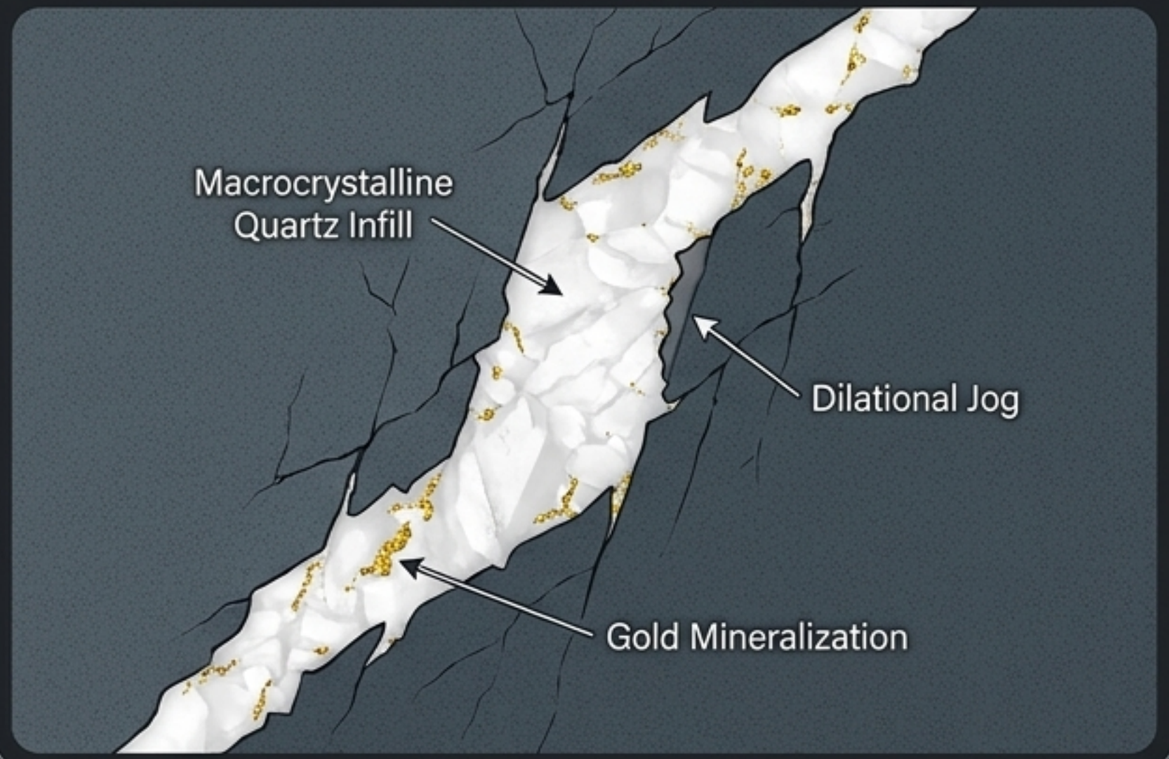
Geological Role: The Fluid Conduit.

Acts as the primary plumbing system for deep-seated hydrothermal, mineralizing, and metamorphic fluids. Fundamental control for lode-gold and base-metal deposits.

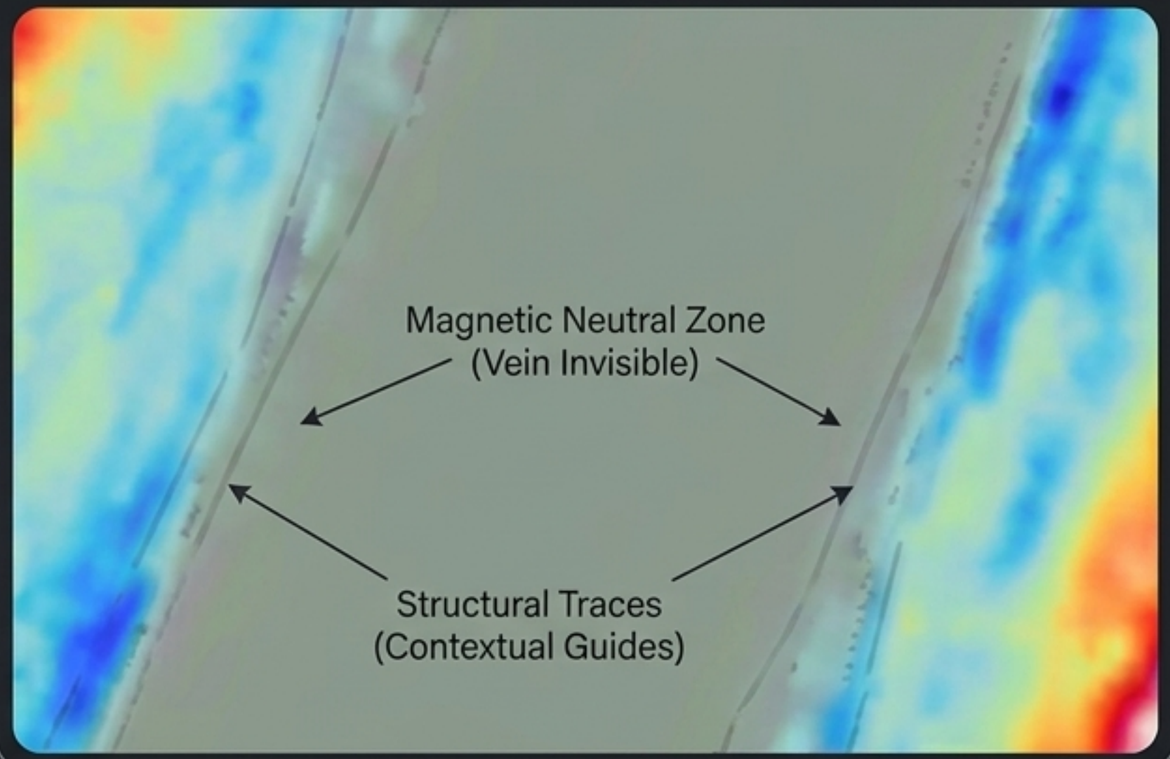
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Feature 3: Quartz Vein

Physical Reality



Magnetic Reality



What it is: Precipitated Silica Infill

A discrete, sheet-like body of macrocrystalline quartz precipitated from silica-rich hydrothermal fluids into open fractures.

Magnetic Expression: Neutral / Invisible

Quartz is diamagnetic. It will not register a positive magnetic response and is generally non-magnetic to weakly magnetic.

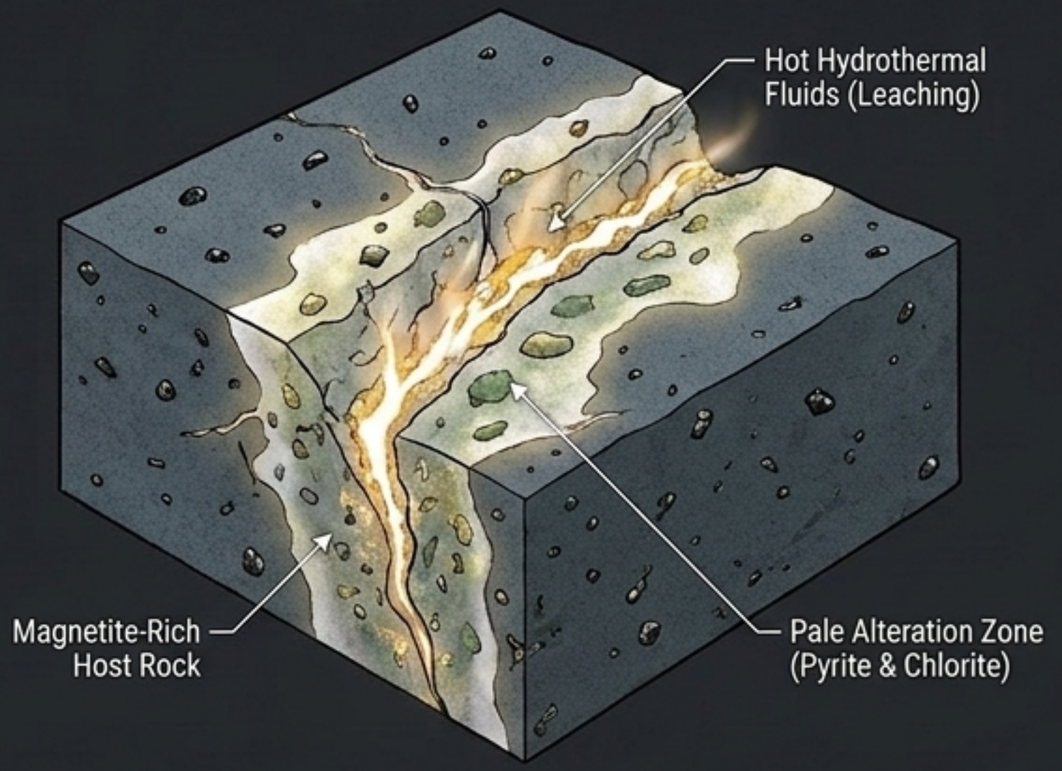
Geological Role: The Mineral Host

Often the direct host for economic mineralization (gold, silver, base metal sulfides), particularly in greenstone-hosted gold deposits.

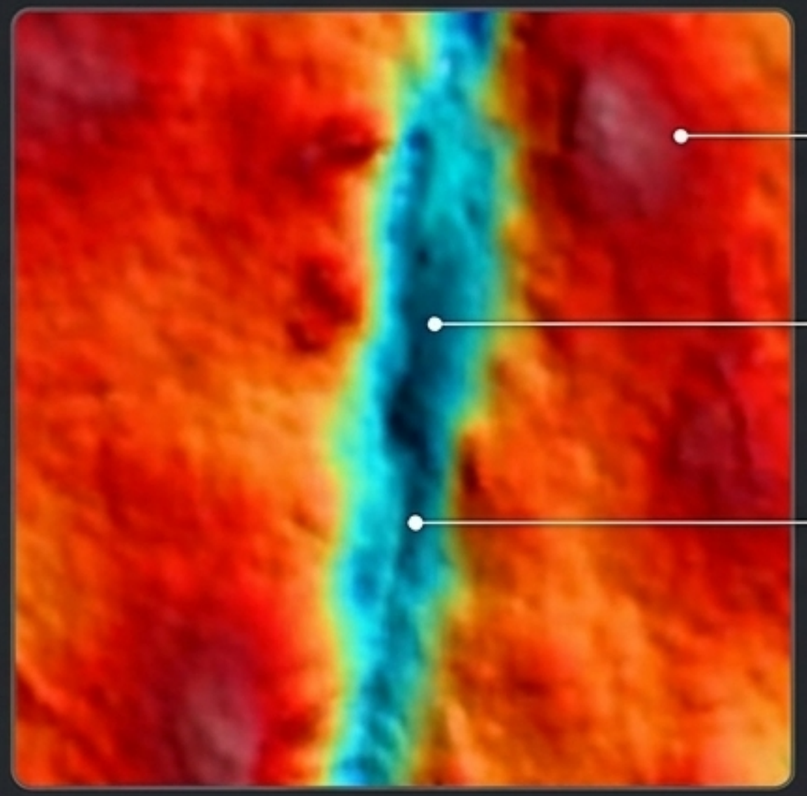
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Feature 4: Magnetite Destruction

Physical Reality



Magnetic Reality



What it is: Chemical Alteration.

Primary magnetic minerals (magnetite) are chemically broken down and replaced by non-magnetic minerals (pyrite, pyrrhotite, hematite) by hot hydrothermal fluids.

Magnetic Expression: The Magnetic Shadow.

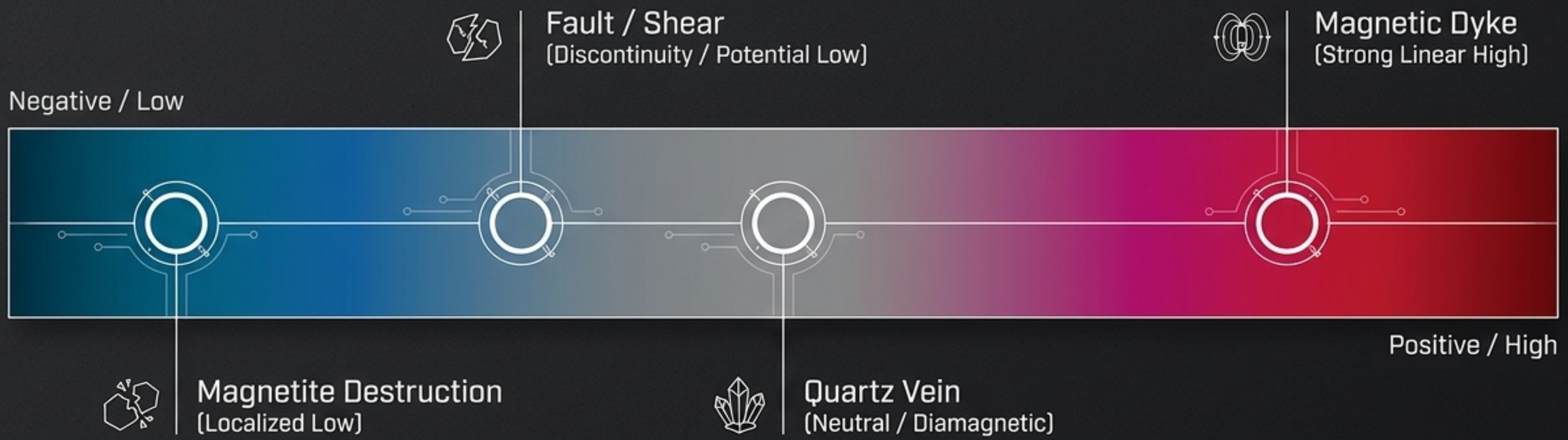
Produces a distinct, localized magnetic low. It is a critical vectoring tool to map zones where fluids have bleached out regional magnetite.

Geological Role: The Indicator.

Direct evidence of fluid-rock interaction, metasomatism, and mineralized alteration envelopes.

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The Magnetic Spectrum



Geophysical interpretation relies on relative contrast. The goal is rarely to find the highest magnetic peak, but to identify the interaction between strong primary rock signatures and the structurally-controlled fluids that destroy them.

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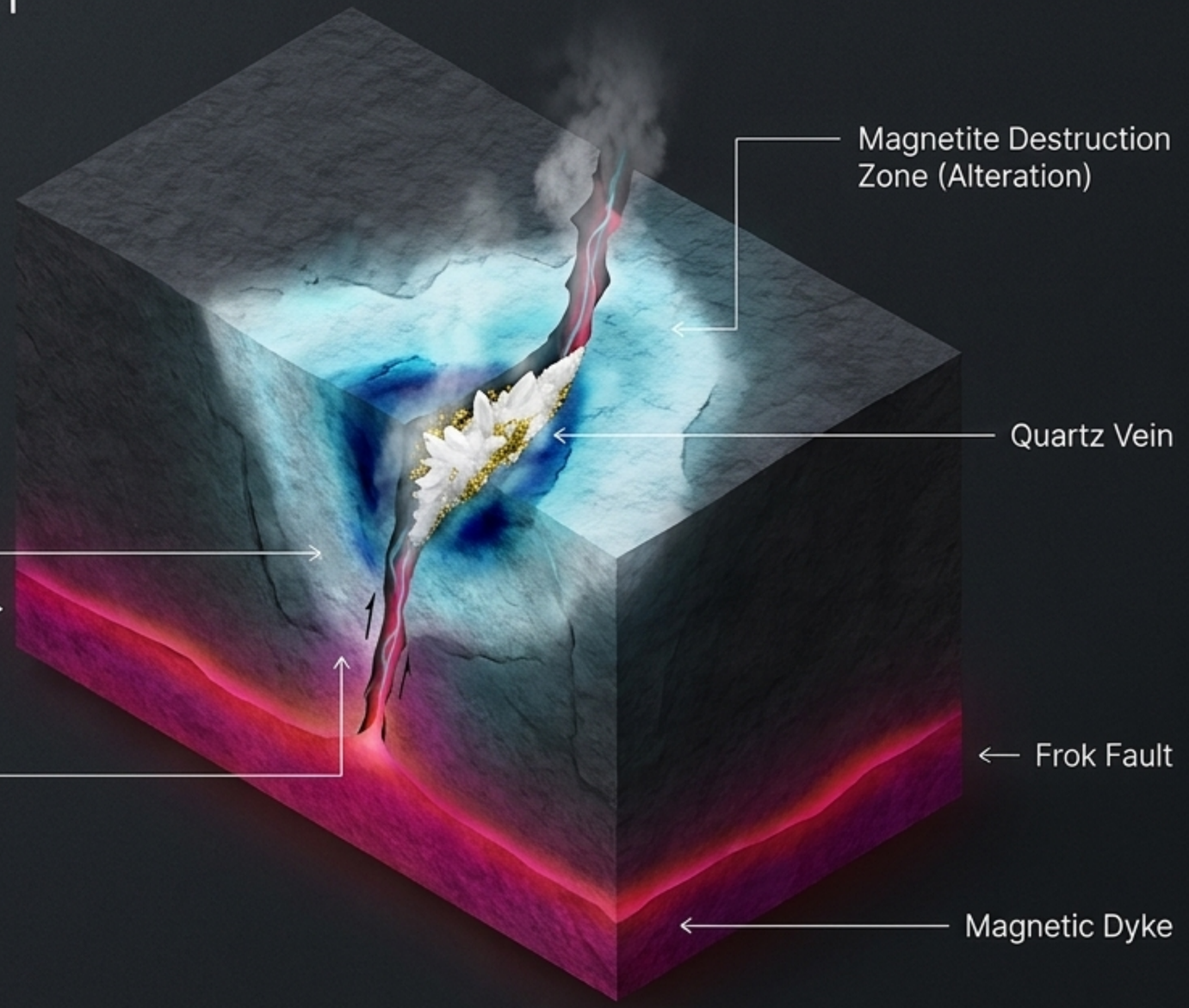
The Unified Mineral System

1: **Thermal Driver.** A magnetic dyke intrudes, providing a heat engine.

2: **Fluid Conduit.** A major fault provides the deep plumbing system.

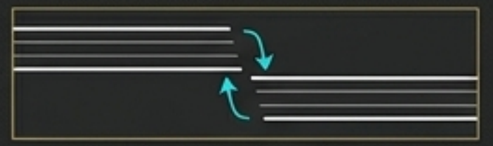
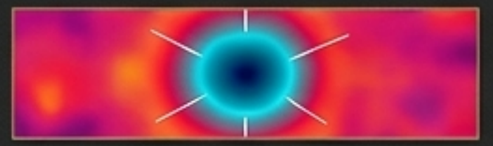
3: **Hydrothermal Alteration.** Fluids travel outward, chemically destroying regional magnetite and leaving a magnetic shadow.

4: **Mineral Deposition.** Silica precipitates in the structural trap, forming a non-magnetic quartz vein hosting gold and sulfides.



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The Diagnostic Matrix

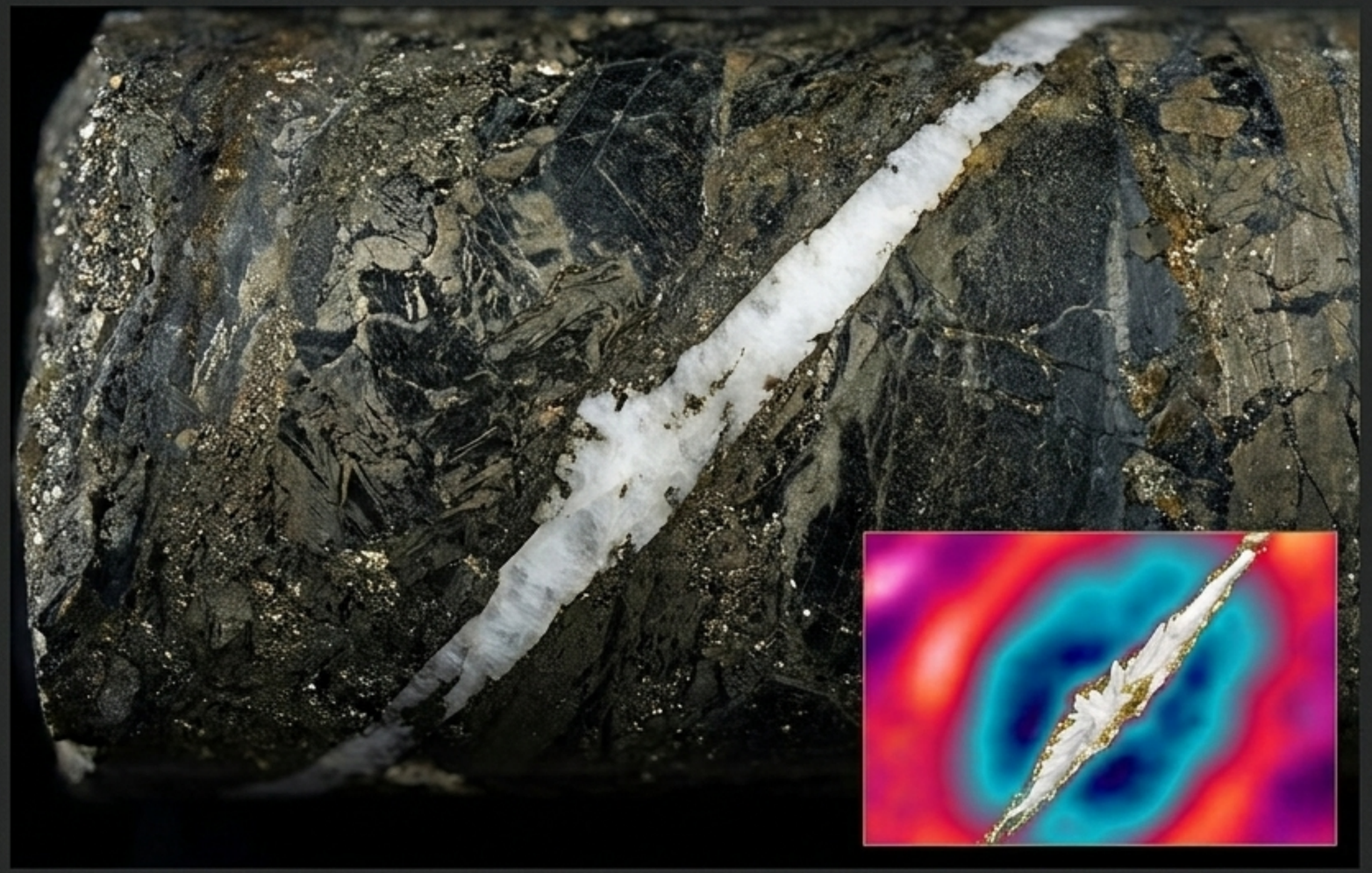
Feature	Physical Nature	Magnetic Signature	Primary Exploration Significance
Magnetic Dyke	Intrusive igneous sheet	 Strong linear High	Structural marker, cross-cutting timeline
Fault / Shear	Fracture / displacement zone	 Linear Discontinuity / Offset	Fluid conduit, structural trap
Quartz Vein	Precipitated silica infill	 Neutral / Non-magnetic	Direct host for gold/sulfide mineralization
Magnetite Destruction	Chemical alteration / fluid reaction	 Localized Low	Hydrothermal alteration halo / fluid pathway

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The Signal in the Stone

Geological mapping and geophysical interpretation are two sides of the same coin.

Mastering the signatures of dykes, faults, veins, and alteration is the key to unlocking the earth's mineral systems.



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